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Diverse classifiers with label dependencies for long-tail relation extraction in big data ☆

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ABSTRACT

Relation extraction is a critical step in knowledge recommendation for big data, but long-tailed distribution in real-world relations presents a significant challenge. Most relations fail to gather enough training instances, which forms a long tail in data distribution and leads to poor performance on these relations. Previous studies have made efforts to improve models for long-tailed relations by sharing knowledge from head classes to the long-tail. Despite proven effectiveness on long tail relations, this line of work lacks control over the knowledge transfer process, which can harm performance on head classes. To address this issue, we propose an approach that enhances the label hierarchical dependencies of a classifier through label-to-sentence attention with multi-granular constraints across different levels of relation. Moreover, We introduce an ensemble mechanism that uses a router module to balance performance between head and tail classes, thereby fixing the long-tail problem on relation extraction data set. Our approach achieves excellent performance on long-tail and all relations on the large-scale benchmark data set New York Times, without sacrificing performance on head relations. The experiment results demonstrate that our approach effectively alleviates long-tail problems and boosts performance on long-tail classes in relation extraction without harming the performance on head classes.

1. Introduction

Information extraction (IE) is widely used in knowledge graphs, information retrieval, question answering systems, sentiment analysis and text mining because it extracts information frames and factual information from big data that users are interested in from natural language. When given a pair of entities in a sentence, relation extraction (RE), as shown in Fig. 1, is aimed at identifying the relationship between the entities, playing a key role in big data utilizing [1] in smart city [2,3]. Structured knowledge, as the outcome of RE, benefits multiple downstream tasks, e.g. knowledge graph construction, information retrieval, malicious traffic detection [4–6], question answering and dialog generation. Researches have put efforts on computational intelligence to improve performance of relation extraction, as a significant step of knowledge recommendation. Supervised models [7] have been proposed for RE tasks and their performance relied on high quality and large scale data. However, the annotation of such data can be highly time-consuming and labor-intensive. [8] applied distant supervision (DS) to automatically generate a labeled training set through aligning entities in knowledge graph to corresponding entity mentions to meet the demand of large-scale data. Dataset from distant

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supervision is on based on the assumption that all sentences containing the same two entities express the same relation containing in the knowledge graph. Since relations are long-tailed distributed in the real world, such automatically generated RE dataset inevitably suffers from long-tail relations.

Long-tail relations has become a limit of improvement and are not to be ignored. On New York Times (NYT) dataset [9–11] , 70% of relations are long-tail, which can be shown in Fig. 2. To mitigate the long-tail problem, researches of sharing information from head relations to tail relations [12–14] are proposed, which took advantage of semantic similarity due to the relation hierarchies of the knowledge graph. Intuitively, attention-based network and its variations in structure put sentence embeddings and label embeddings relation into interaction to score a pair of entities at the level of a sentence bag, to help alleviate wrong-labeling problem and long-tail problem in DS datasets.

Most of the researches fixing on long-tail relations adopt Convolutional Neural Network (CNN) structures for feature extraction and relation representation, regardless the global information contained in the input sequence. By contrast, transformer [15] is able to simultaneously perceive the global information contained in the input sequence and handle more complex contexts. With powerful sequence modeling capabilities, transformer as feature extractor improves performance for downstream tasks, e.g relation extraction specifically, allowing models to generate more accurate relation representation.

In this paper, we present a novel neural model for relation extraction for large-scale data, introduce constrain on label dependency into sentence-to-relation structure and utilize expert-router mechanism to mitigate the long-tail problem. We adopt transformer-based model Bert [16] for the external semantic information. The attention-based network with certain constrain utilize information in relation labels, which shows better performance on single classifier. The overall performance of the general model is boosted by training multiple variant expert classifiers with a distribution loss, followed by a router layer that integrates outcomes from all experts and select the most possible combination.

Our model show effectiveness especially effective on datasets with multi-granular relations. We use five objectives to jointly train our proposed model. The first objective optimizes the attention weights of relation at multi-granular level to achieve semantic augmentation. The second is aimed at predicting the relation label at the level of a sentence bag, which can be regarded as the main goal of relation extraction tasks. The third objective minimizes difference between higher level relations and the mean of lower level relations served as key and value for attention which optimizes the representation of labels and enhance the relation hierarchy. The fourth objective is a distribution loss that maximize the different between different expert classifiers and encourages them to generate complementary results. The last objective is to select the most convincing results from combinations of experts.

Our experimental results on large-scale NYT dataset reflect our contributions, which lie in two parts:

1. Our label-to-sentence attention model with multi-granular constrain among different level of relations utilizes the hierarchy of relation labels. The Bert-based relation extractor introduce outer information into relations, while our multi-granular constrain enhance label hierarchical dependencies for more accurate representation of relation labels.

2. We take advantage of an ensemble mechanism with certain adaption made for relation extraction. The application of the expert-router mechanism balance the performance between head relations and tail relations. As far as we know, the expert-router is not yet appeared in relation extraction field.

3. The results show that our model boost performance on both head relations and tail relations.

This paper includes five sections. The first section is the introduction of the research background. The second section describes related work. The third section demonstrate our model in detail, while the fourth section performs experiments and proves effectiveness. The last section is the conclusion for this paper.

2. Related works

2.1. Long-tail approaches for relation extraction

Large scale of annotated data for supervised RE models training is demanded, which is highly time-consuming. To overcome this problem, [8] utilized distant supervision to annotate data automatically. Distant supervision inevitably brings in wrong labeling samples. To overcome noisy samples in the dataset, [10] brought forward multi-instance learning (MIL) methods. Due to the development of language models, researchers employ neural models for RE to capture hidden textual information for relations instead of linguistic analysis [17,18]. To push RE further, some studies incorporate external textual information [19,20] and advanced training strategies. [21] exploited the dependency between words and the entities by pre-trained language models. These works mainly focused on the wrong-labeled data caused by DS in large-scale datasets, regardless of almost most relations in the datasets being long-tail relations.

Only a few researches focus on long-tail relations for RE [11,12,22]. Of these approaches, [22] proposed an explanation-based approach, whereas [11] used a rule-based method, utilizing logic rules as external knowledge. In these studies, relation are not considered connected, despite the semantic correlations among the relations. Because of the effectiveness of selective attention [17] in multi-instance learning, recent work applied selective attention to form a baseline and then proposes specific approaches for mitigating wrong-labeling or long-tail problem [17] by utilizing high-level relation embeddings along with low level relations to obtain more relation information. Namely, high level relation embeddings perform only as queries of selective attention to generate information at the level of a sentence bag. [12] put forward a hierarchical attention structure specifically for long-tail relation extraction. [13] uses graph embeddings to share the information from relations with adequate data to the long-tail relations. The sharing is based on the semantic correlation between relations, which exists in the relation hierarchies of a knowledge graph, i.e. Freebase. [12] use end-to-end structure and randomly initialize relation embeddings, while [13] concatenate static TransE [23]

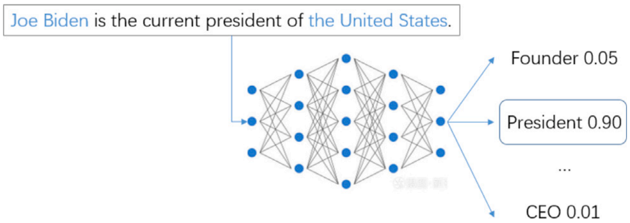

Fig. 1. An example of RE models. When given two entities and one sentence mentioning them, the relation of two entities is classified by RE models within a pre-defined relation set.

embeddings pre-trained on Freebase with and dynamic GCN [24] embeddings according to the hierarchies of relation labels. Despite this approach manage to alleviate long-tail problem, the generation of graph embeddings cost highly and lack practicability. [14] utilize the weight of sentence-to-relation attention, which is put into classification lost as an optimization for the above approaches.

These works are proven to improve performance on long-tail relations, whereas the above models have two issues: (1) The hierarchy information of relation labels have not been well exploited. (2) Single classifier lack variety of the output, leading to greater variance of results. For the above problems, we propose a constrain to enhance hierarchy of relations and introduce router-expert mechanism in image classification field for long-tail labels.

2.2. Long-tail approaches for image classification

Researches for long-tail data in image classification field is more diversified in the past few years. Researchers put efforts in few-shot learning, re-weighting and knowledge transfer. Among these approaches, few-shot learning relies on balanced data while long-tailed data is highly imbalanced, while re-weighting cause instability to tail classes and knowledge transfer cause performance drop on head classes. To overcome the above issues, irrelevant to attention network, other researches focus more on ensemble and grouping. The samples are broadly divided into head groups and long-tail groups, following by a single model with experts concentrating on each group to generate a multi-expert structure. BBN [25] uses two branches, one focuses on the head classes while the other takes notice of the long-tail. LFME [26] learns from several teacher models, each focuses on classes with different quantity of samples and distills a model from the teachers. Though performance is improved at tail classes, BBN and LFME still cannot escape from performance loss at head classes. [27] proposed Routing Diverse Experts (RIDE), which is a non-traditional ensemble method. It reduces the model variance for all the classes and significantly reduces the model bias for the tail classes. Besides, it also increases the mean accuracies for all class splits in image datasets such as CIFAR and ImageNet. (1) Its classifiers share the same early layers and are jointly optimized to reduce small tail overfitting. (2) It dynamically deploys classifiers with a router module.

Different from other RE approaches for long-tail problem, we combine mechanism similar to these multi-expert approaches with label attention to solve long-tail in RE, without harming the performance of the head relations.

3. Routing classifiers with label dependencies

A illustration of our intelligent model Routing Classifiers with Label Dependencies (RCLD) is shown in 3. A sentence bag with the same head and tail relations is tokenized and converted into relation representation. Next, the relation representation is put into multiple novel label attention classifiers with label dependency for each classifier and diversity loss between them. Lastly, the router generates results from the output of multiple diverse classifiers. In this section, we begin with the definition of relation extraction of a sentence bag for big data. Next, we introduce constrain on relation labels to obtain hierarchy information. Lastly, we provide routing diverse expert which include multiple expert classifier and an attention-based router that select the most possible output. We present an illustration of the model in Fig. 3, while detail of part of the model is shown in Fig. 4.

A sentence bag is formed by sentences with the same pair of head and tail entities, given a relation label automatically by DS. Relation extraction for DS datasets aims to predict the predefined relation label under the condition of a sentence bag with head tail entities specified. Relation labels are hierarchical, allowing high level relation, e.g. coarse-grained relations, to pass information top down.

First, we utilize one of the Bert representations as the sentence-Level representation for a relation. In order to obtain semantic information from outer corpus, we apply a Bert-based extractor as the feature extractor for sentences, while the relation representation is generated as done in [28], who take the start marker of both entities and use their mean as the output, which beat other representation so far as Bert representation is concerned. Namely, a single sentence is tokenized to a sequence of tokens, followed by feature extractor turning these tokens to low-dimensional and dense word embeddings.

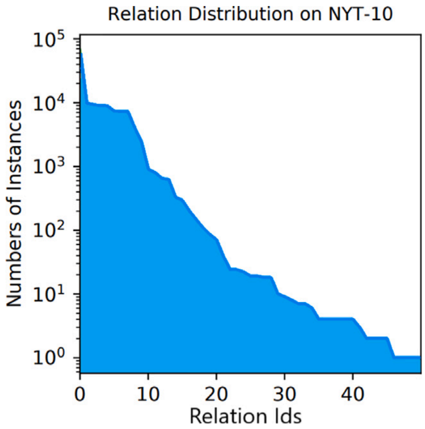

Fig. 2. Relation distributions (log-scale) on the training part of DS datasets NYT-10 and Wiki-Distant, suggesting that relation distributions in real-world are faced with the long-tail problem.

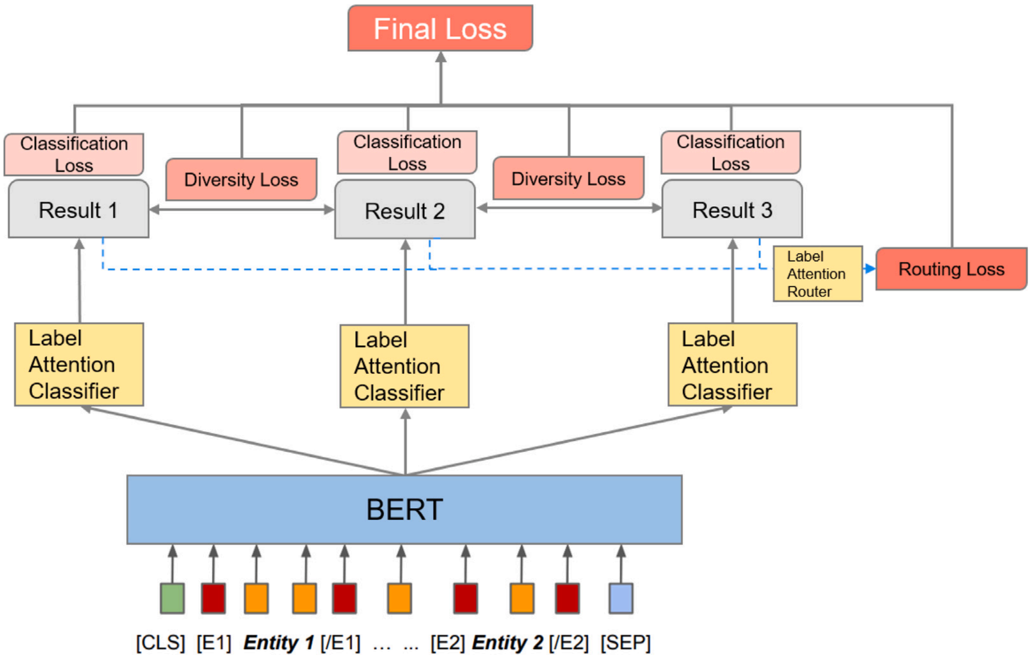

Fig. 3. An illustration of our proposed model.

3.1. Multi-granular dependency attention

Our model inherited the sentence-to-relation (sent2rel) attention in [14], which aims at augmenting sentence representation with relation embeddings. The sentence representation $\boldsymbol{s}$ functions as a query to multiply the relation embedding matrix $\boldsymbol{R}^{(0)}$ as a key and a value by a linear layer.

$$\boldsymbol{\alpha}^{(0)} = \mathrm{softmax}\left(\boldsymbol{s}^T \boldsymbol{R}^{(0)}\right) \tag{1}$$

$$\boldsymbol{c}^{(0)} = \boldsymbol{R}^{(0)}\boldsymbol{\alpha} \tag{2}$$

where $\boldsymbol{c}^{(0)}$ is the resulting relation-aware representation corresponding to the sentence $\boldsymbol{s}$. On this basis, for each higher level relation with multiple children, we consider the semantic connection between a parent and all its children. We formulate a multi-granular constrain that help construct a more compact hierarchy, meaning a higher level relation should be close to the average of lower level relations. We calculate the mean of the lower level relation representation in the same higher level class, then take the higher level relation representation. Cosine embedding loss is used for measuring whether two inputs are similar or dissimilar, using the cosine

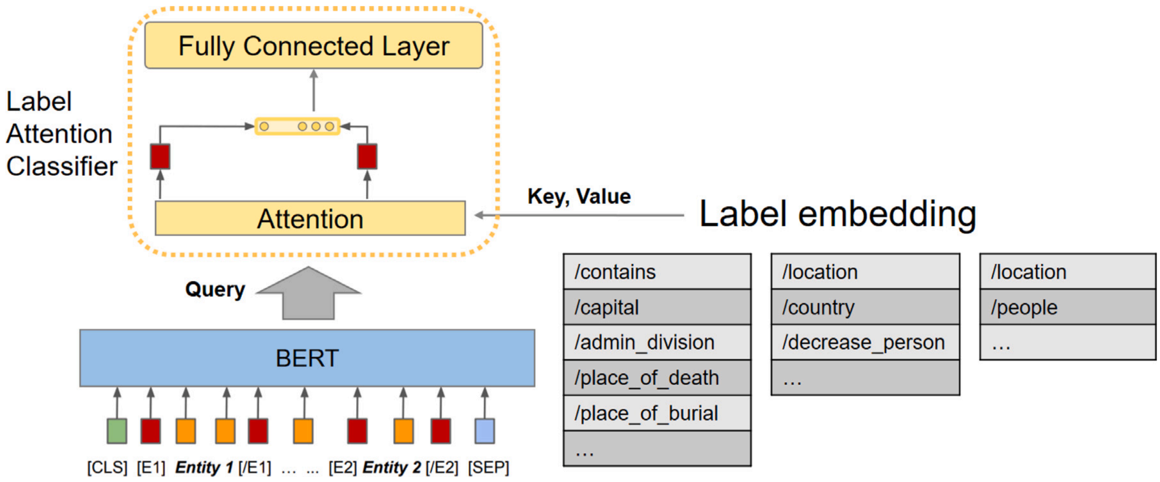

Fig. 4. A detailed figure of the label attention classifier in our proposed model. The attention router is similar as above.

distance, and is typically used for learning nonlinear embeddings or semi-supervised learning. We minimize a cosine embedding loss to achieve the above goal.

$$\mathcal{L}_{Cosine} = \mathcal{L}_{Cosine1-2} + \mathcal{L}_{Cosine2-3} \tag{3}$$

while $\mathcal{L}_{Cosine1-2}$ is the cosine embedding loss between relation level one and two, besides $\mathcal{L}_{Cosine2-3}$ is the cosine embedding loss between relation level two and three.

3.2. Routing diverse distribution-aware experts

[27] brought forward a multi-expert method which shared earlier layers f_θ and divided n uncorrelated later layers $\Psi = \{\psi\theta_1, \dots, \psi\theta_n\}$. At the first step, these experts are jointly optimized. Next, a learned router module is used to dynamically assign these experts at the second step. During the inference stage, all logits of the m selected experts are averaged for the final softmax classification. If individual experts makes independent decisions, softmax of the average logits represent the product of individual classification probabilities, which approximates the joint probability of experts.

Consider n independent experts of the same sentence-to-relation attention structure, we share the same earlier layers f_θ for all the n experts. Experts keep uncorrelated later layers $\psi\theta_i$, $i = 1, \dots, n$. All n experts are optimized at the same time on long-tailed data distribution-aware diversity loss $\mathcal{L}_{D-Diversify}$ and individual classification loss $\mathcal{L}_{Classify}$, i.e. Cross Entropy Loss.

Distribution-aware diversity loss. The separate classification loss, along with randomly initializing can encourage expert classifiers to be diversified. As a complement, the Kullback–Leibler divergence, a measure of differences between two probability distributions, P and Q. A regularization term is added to promote complementary results from experts for better performance of long tail data. That is, we maximize the Kullback–Leibler divergence between the logits among different experts on instance x with label y in c classes.

Routing diversified experts. To cut down computational cost caused by multiple experts, we train a router to generate results from the sequential expert classifiers. Assume that the first k experts are selected for instance x. The relation feature and the mean logits from the first to the kth expert are input to the router, which generates a classification result y_{on} on whether to activate the $k+1$-th expert. When the kth expert gets incorrect result for x, but one of the rest $n-k$ experts gets the correct result for x, the router should be turned on, i.e., output $y_{on} = 2$, and otherwise $y_{on} = 1$ if expert 1 correctly classifies and expert 2 does not. We construct a attention classifier with an attention layer and a fully connected layers and utilize the attention weights. Specifically, we attend relation embeddings to the relation feature, followed by a linear layer $\boldsymbol{W}$ with activation to reduce the dimension of relation features. We formulate a k-class cross-entropy loss for the router layer, denoted as $\mathcal{L}_{Router}$. At the test time, we simply argmax the logits of the router layer: If the logit of expert k is the max logit, the classifier gets the final result with the average of front k logits.

According to above, the final loss of the expert-router module should be:

$$\mathcal{L}_{expert-router} = \mathcal{L}_{D-Diversify} + \mathcal{L}_{Classify} + \mathcal{L}_{Router} \tag{4}$$

4. Experiments

This section demonstrate results of experiment of our proposed intelligent model for big data. We conduct evaluation of the proposed model on benchmark dataset with multiple standards.

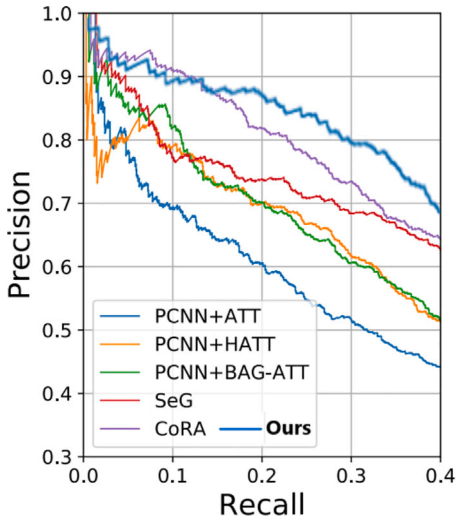

Fig. 5. Precision–recall (PR) curves on NYT-10.

Table 1
Results of model Evaluation on NYT using P@N and AUC.

P@N(%)	100	200	300	Mean	AUC
CNN+ATT [17]	74.3	71.5	64.5	70.1	0.35
PCNN+ATT [17]	76.2	73.1	67.4	72.2	0.39
PCNN+HATT [12]	88.0	79.5	75.3	80.9	0.42
PCNN+BAG-ATT [29]	91.8	84.0	78.7	84.8	0.42
SeG [30]	93.0	90.0	86.0	89.3	0.51
CoRA [14]	98.0	92.5	88.3	92.9	0.53
Ours	97.2	91.7	89.4	92.1	0.54

Dataset and evaluation metrics. As done in works of DS relation extraction [12,17], we continue to use New York Times (NYT) dataset [9], benchmark of DS relation extraction, to evaluate our model. It contains 53 different relations, where NA (None of the above) class indicating that the relation is not predefined in the relation set. The NYT dataset consists of 570,000 samples of sentence in training set and 172,000 in test set. Top-n precision (P@N), along with precision–recall curve and area under curve (AUC) and are employed as measurements of the performance of a relation extraction model. Hits@K is also evaluated for long-tail relations by following [13]. P@N (top-n precision) is the precision values for samples with top-100, top-200 and top-300 confidences of positives, while Hits@K denotes whether top-K prediction inferred from confidence of a sample matches the ground truth.

Comparative approach. Our approach is compared with the following previous works. PCNN+ATT [17], PCNN+HATT [12], PCNN+BAG-ATT [29], PCNN+KATT [13], SeG [30] and CoRA [14].

4.1. Evaluation on benchmark

Shown in Table 1, our proposed model is compared with previous approaches on relation extraction benchmark on top-n precision, PR curve and AUC. Results on these metric show superior performance of our proposed model compared to other baselines. It also surpasses other baselines on long-tail relations as in Fig. 2.

In terms of P@N, our proposed model significantly improve the performance by 25% on P@100, 25% on P@200 and 25% on P@300 compared to CNN with selective attention (e.g., CNN+ATT [17]). In addition, compared to state-of-the-art models with single classifier methods, i.e. CoRA [14] that also utilizes relation hierarchies, our proposed model manages to achieve better results in AUC and P@300, at the cost of only a tiny drop on P@100 and p@200 by 0.8. We also present Precision–Recall curves in 5. The curves shows that our proposed method present larger area under P-R curve compared to other models. The AUC metric show that our model improves AUC by 0.01 and reaches a new state-of-the-art performance by using multi-granular dependency attention and expert-router mechanism. The curve of our model brings a performance promotion on precision when recall falls in the area between 0.1 and 0.4. Specifically, our model improves both PCNN+HATT and PCNN+BAG-ATT by 28.4% in terms of AUC for precision–recall.

4.2. Ablation study and evaluation on long-tail relations

Ablation study, shown in Table 2 is conducted to further verify the effectiveness of both our proposed modules. On one hand, we replace the expert-router module with single linear classifier. As a result, compared to our proposed model, base model without

Table 2
Model ablation study on NYT.

P@N(%)	100	200	300	Mean
Ours	95.2	91.7	89.4	92.1
Ours w/o label dependencies	93.2	91.2	90.1	91.5
Ours w/o multiple experts	94.9	91.4	89.1	91.8

Table 3
Hits@K (Macro) on the relations whose number of training instance < 100/200, macro meaning taking average Hits@K for all classes regardless of number of samples in each relation.

Training Instances	<100			<200		
Hits@K (Macro)	10	15	20	10	15	20
PCNN+ATT [17]	<5.0	7.4	40.7	17.2	24.2	51.5
PCNN+HATT [12]	29.6	51.9	61.1	41.4	60.6	68.2
PCNN+KATT [13]	35.3	62.4	65.1	43.2	61.3	69.2
CoRA [14]	66.6	72.0	87.0	72.7	77.3	89.4
Ours	64.3	72.3	88.2	74.2	79.2	90.4
Ours w/o label dependencies	62.7	70.8	86.5	72.3	76.4	87.4
Ours w/o multiple experts	61.5	69.8	86.1	70.1	77.1	86.8

expert-router module shows an obvious drop on long-tail relations by about 2 on Hits@K while the number of training instances is below 100 and 200, as detailed in next paragraph. Model without the expert-router module also shows and a minor gap in terms of P@N. On the other hand, we simply remove the dependency loss to verify the importance of the multi-granular dependency constrain in training process. Results shows that removing the constrain among levels of relation labels also causes a mild drop on all relations. The above results prove that label dependencies and expert-router module both play an important role in our proposed approach.

To prove the superiority of our proposed model in dealing with long-tail relations, we evaluated our model specifically on long-tail relations. The same evaluation setting as [12,13], Macro Hits@K is used to reflect performance on long-tail relations. As shown in Table 3, we compare our proposed model with others. Results on Hits@K shows that our proposed model improves the performance on long-tail relations by a large margin compared to selective attention methods. Even compared to state-of-the-art results, i.e. CoRA [14], our model achieve better results on Hits@K under the circumstance where training instances is less than 200. The performance gap between our model and others demonstrates the advantages of our framework in handling long-tail relations.

In the aspect of ablation study on long-tail relations. As shown in the last three rows in 3, model without expert-router module leads to a performance loss by about 2.8 on Hits@10, 2.5 on Hits@15 and 2.1 on Hits@20 when training instances are less than 100, on the other hand 4.1 on Hits@10, 2.1 on Hits@15 and 3.6 on Hits@20 when training instances are less than 200. Though the margin is not as large as above, model without dependency loss leads to quite a performance gap on all statistics at about 2.0 on each metric, which also emphasizes its importance for long-tail relations.

5. Conclusion

We propose an intelligent model with novel mechanism to meet the demands of long-tail data in relation extraction, as one of the most important steps in knowledge recommendation for big data. Recent work focusing on long tail problem in relation extraction put efforts on knowledge transfer between head and tail relations, which harm the performance on head relations. Single classifier in selective attention methods also lack capability to learn from long-tail relations. To mitigate long-tail problem in relation extraction, we bring forward a relation hierarchy dependency that exploits semantic information among relation labels. The hierarchy dependency is next integrated into the label-to-sentence attention as a constrain for knowledge transfer between head and tail relations. To fix the insufficient capacity for learning long-tail relations of single classifier, We apply an ensemble mechanism including multiple classifiers that focusing on different relations and a router layer that selects the best result combination of all classifiers. To prove the effectiveness of modules in our proposed model, We conduct experiments on large scale relation extraction benchmark data set New York Times using both normal relation extraction metrics (e.g., PR curve, AUC and top-n precision) and long-tail metrics (e.g., Hits@K). Results shows that our proposed approach reach excellent performance on dealing with both head relations and long-tail relations.

Despite the capability to improve overall performance, deficiencies on our proposed model are not to be ignored. On one hand, we apply Bert as feature extractor for better relation representation, causing high time complexity in training and inferring process. On the other hand, our proposed model focuses on long-tail problem in relation extraction, without paying attention on alleviating wrong labeling problem that could further improves overall performance. For further investigation, we will make efforts on following aspects: (1) reducing time complexity of the model without harming performance. (2) investigating more complex attention structures to reduce the impact of wrong labeling problem caused by distant supervision.

Declaration of competing interest

All authors declare that there it no conflict of interest.

Data availability

Data will be made available on request.

Acknowledgments

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